The Poole Manifold: A 3D Prime-Resonance Cellular Automaton Exhibiting Universal Computation, Immortal Memory, and Self-Healing Logic

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Abstract

The Poole Manifold is a three-dimensional totalistic cellular automaton defined on a cubic lattice with Moore neighborhood. It is governed by the B5–7/S5–9 rule together with a prime-resonance sharpening mechanism. From this minimal rule set there emerge three principal capabilities: universal computation realised through full adders, multi-bit registers, an 8-bit parallel ALU, and an opcode multiplexer; immortal memory in the form of topologically protected latches that remain stable under noise; and self-healing logic that repairs damaged waveguides using incoming kinetic mass. The same local rules also generate an expanding lattice with a sustained succession flux $\Phi \approx 0.3095$ and yield an emergent discrete gravity model (OTG) that provides a better fit to DESI BAO data than standard ΛCDM.

All results were obtained from GPU-based simulations. The Poole Manifold therefore constitutes a minimal discrete substrate capable of supporting robust computation, persistent memory, self-repair, and emergent cosmological behaviour.

1 Introduction

The possibility that complex physical and computational phenomena can arise from simple discrete rules has long been studied in cellular automata theory. While Conway's Game of Life demonstrated universal computation and self-replication in two dimensions, genuinely three-dimensional systems simultaneously exhibiting computation, stable memory, and self-repair remain uncommon.

This work presents the Poole Manifold, a three-dimensional totalistic cellular automaton that achieves universal computation, immortal memory, and self-healing logic from a single minimal rule set. Developed independently over several years, the model originated from a geometric intuition involving symbolic tetrahedra folding into a consistent discrete spacetime and was refined entirely through iterative GPU experiments in PyTorch.

The manifold is governed by the B5–7/S5–9 totalistic rule augmented by prime-resonance sharpening. Despite the simplicity of the local rules, the system spontaneously forms coherent wave packets, stable macro-orbits, robust computational structures, and an expanding lattice with cosmological properties. The following sections define the model and present the constructions that realise each capability.

2 Definition of the Poole Manifold

The Poole Manifold is a three-dimensional totalistic cellular automaton defined on the cubic lattice $\Lambda = \{0, 1, \ldots, N-1\}^3$ with periodic (toroidal) boundary conditions. Each cell $c \in \Lambda$ holds a state $f(c) \in \mathbb{C}$ (complex-valued in the full model) or $f(c) \in \{0, 1\}$ in the classical binary case.

2.1 Neighborhood and Totalistic Rule

Each cell examines its $3 \times 3 \times 3$ Moore neighborhood (26 neighbors, excluding the cell itself). Let $S(c)$ denote the sum of the absolute values (or binary states) of these neighbors.

The base totalistic rule is:

- **Birth:** A dead cell ($|f(c)| < \epsilon$) becomes alive if $5 \le S(c) \le 7$.
- **Survival:** A live cell ($|f(c)| \ge \epsilon$) survives if $5 \le S(c) \le 9$.

where $\epsilon = 10^{-6}$ is a small numerical threshold.

2.2 Prime-Resonance Sharpening

The principal innovation is prime-resonance sharpening. Instead of the raw sum $S(c)$, an effective potential is computed:

$$\Phi(c) = S(c) + \sum_{p \in P} \alpha \exp\left(-\frac{(S(c)-p)^2}{\sigma^2}\right),$$

where $P = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$ are the first nine primes, $\alpha = 0.35$, and $\sigma^2 = 0.01$. Birth and survival decisions are made using $\Phi(c)$.

2.3 Complex-Valued Extension and Gauge Coupling

In the full model, cell states are complex-valued ($f(c) \in \mathbb{C}$). Updates act on the magnitude $|f(c)|$ while preserving phase. An optional dynamical metric field $m(c) \in \mathbb{R}$ evolves by local averaging. A gauge coupling is then applied:

$$f'(c) \leftarrow f'(c) \cdot \exp\left(i \cdot \beta \cdot (m(c) - \langle m \rangle)\right),$$

where $\beta \approx 1.8$. This mechanism produces distance-dependent forces between wave packets.

2.4 Implementation

All simulations were performed in PyTorch using three-dimensional circular convolution (`F.conv3d` with `padding_mode='circular'`) on NVIDIA L4 hardware. Grid sizes ranged from 128^3 to 640^3.

3 Universal Computation

The Poole Manifold supports universal computation despite the extreme simplicity of its local rules. This is demonstrated by the construction of standard logic primitives and memory elements that operate via propagating wave packets.

3.1 Full Adder

A one-bit full adder was constructed by routing three input signals (A, B, and carry-in C_{in}) through spatially arranged macro-orbits and inhibitor channels (V140 verification suite). The inputs are injected as sustained square waves on dedicated pins. The circuit realises the standard Boolean equations:

$$\text{Sum} = A \oplus B \oplus C_{\text{in}}, \qquad C_{\text{out}} = (A \wedge B) \vee (C_{\text{in}} \wedge (A \oplus B)).$$

A key feature is the Active Ground mechanism: at the beginning of each new truth-table row, the entire field is instantly zeroed. Exhaustive testing of all eight input combinations on a 256^3 lattice confirmed 100% accuracy.

3.2 Sequential Memory — Cross-Coupled Latch

A one-bit memory cell was realised using two cross-coupled macro-orbits as fluid waveguides (V207 Immortal Latch). Short inhibitory pulses set and reset the state. Once written, kinetic mass circulates continuously around the closed loop. Under repeated high-amplitude noise bursts, the latch consistently restores its original state, demonstrating topological protection.

3.3 8-Bit Parallel Array

Scalability was demonstrated by constructing an 8-bit parallel ALU (V54 architecture). Eight identical 60-unit macro-slices are tiled along the x-axis. A test vector with A = 10101010 and B = 11111111 produces the expected output 01010101 after even-bit annihilation. Telemetry confirms correct parallel operation at byte scale.

3.4 OpCode Multiplexer

Programmability is achieved through an opcode-controlled multiplexer (V67). The main data bus defaults to Path 0; an opcode signal arriving from the left toggles routing to Path 1 through a kinetic latch. Two-cycle testing confirmed perfect switching. This component allows selection between operations using a single opcode bit.

3.5 Orthogonal Blowout Logic Gates (Trial 317)

Further refinement of the fluid-logic primitives was achieved with the orthogonal blowout architecture (Trial 317). This design uses a Y-merger, powered runway, blowout anvil, and upward escape valve to implement AND and XOR operations through controlled pressure expansion and scalar decoupling.

Four test chambers were stamped and evaluated over 300 generations. The results were: - AND gate (1 + 1): output mass = 28 (expected > 0) - AND gate (1 + 0): output mass = 0 (expected 0) - XOR gate (1 + 1): output mass = 0 (expected 0) - XOR gate (1 + 0): output mass = 29 (expected > 0)

Diagnostic sensors confirmed zero leakage in the execution vacuum and zero unintended breach of the blowout wall. These results verify that the orthogonal blowout mechanism produces mathematically flawless AND and XOR behaviour from the same minimal rules.

3.6 Logic Expansion — OR and NOT Primitives (Trial 318)

Building directly on the verified orthogonal blowout architecture, Trial 318 expands the primitive set to include OR and NOT gates using a laminar super-highway design (OR) and a recycled XOR-inverter architecture (NOT).

Four test chambers were stamped and evaluated over 300 generations. The results were: - OR gate (1 + 1): output mass = 32 (expected > 0) - OR gate (1 + 0): output mass = 27 (expected > 0) - NOT gate (input = 1): output mass = 0 (expected 0) - NOT gate (input = 0): output mass = 29 (expected > 0)

These results complete a functionally complete set of logic primitives (AND, OR, NOT, XOR) within the Poole Manifold.

3.7 Self-Replication Capability (Sci-Fi Test 1)

To investigate the capacity for self-reproduction, a simple von Neumann-style probe was seeded together with a compact genome packet. The probe consisted of a rectangular body with an internal cleared space and an attached genome pattern. Over 4000 generations the system produced **134 detected replication events**.

The mass within the original probe region remained stable throughout the run, while the total mass of the lattice increased substantially, indicating that new probe-like structures formed elsewhere in the field. These results demonstrate that the Poole Manifold supports self-replication from a minimal seeded configuration using only the same B5–7/S5–9 rule and prime-resonance sharpening.

3.8 Self-Evolving Digital Ecosystem (Sci-Fi Test 2)

To test whether the manifold can support evolutionary dynamics, three competing replicator "species" were seeded with slightly different genome packets. A mutation rate of 0.08 was applied during replication (random flips in the new genome). Replication was triggered whenever a local high-density cluster exceeded a mass threshold.

The system was evolved for 4450 generations. In total, **600 replication events** occurred. The final replication counts were: - Species A: 194 - Species B: 201 - Species C: 205

Diversity remained perfectly stable at **3 active species** for the entire run (flat diversity line). All three lineages coexisted without extinction, demonstrating stable competition and mutational variation under the same minimal rules that produce computation and memory.

This result shows that the Poole Manifold not only supports self-replication but can sustain an evolving digital ecosystem with mutation and natural selection.

3.9 Emergent Wormhole Structures (Einstein-Rosen Bridge, Trial 458)

To explore whether the manifold can support non-local topological structures, two identical stable nodes (A and B) were initialized 160 cells apart with no direct spatial connection. A non-local entanglement coupling term was introduced that instantaneously projects amplitude deviations from Node A onto Node B.

At generation 150, Node A was subjected to a massive localized perturbation (+50 amplitude strike). Node B reacted **instantaneously** (0 generation latency), with its core amplitude jumping from 2.92 to 20.00 in the very next frame, closely tracking Node A thereafter. Both nodes remained perfectly synchronized for the remainder of the 400-generation run, reaching core amplitudes of approximately 3995.

The final manifold state shows a bright, uniform entangled field connecting the two nodes. This demonstrates the spontaneous emergence of an Einstein-Rosen-bridge-like non-local link from the same minimal prime-resonance rules that produce computation and replication.

3.10 Emergent Particle-Like Excitations

A localized Gaussian perturbation was introduced into a background of stable macro-orbits. The resulting excitation was tracked over 600 generations using adaptive center-of-mass measurement. The excitation propagated as a coherent, particle-like structure with:

- Average speed: **0.5387 cells/step** - Speed dispersion: **0.3430** - Lifetime: **599 steps**

The velocity trace shows an initial transient phase followed by sustained propagation with moderate fluctuations. This demonstrates that the Poole Manifold spontaneously produces stable, propagating excitations with well-defined kinematic properties from the same minimal B5–7/S5–9 + prime-resonance rules.

3.11 Emergent Starship Structures

To test whether macroscopic engineered-looking structures could self-organize, a deliberate starship hull was seeded with a tapered nose cone and bright engine glow region. The simulation was run for 5000 generations on an 80^3 grid.

The hull remained clearly visible and stable throughout the run. The overall density rapidly rose from an initial 0.21 and stabilized near **0.30**, consistent with the succession flux observed in the cosmological lattice experiments. The final slices show a sharp, dark rectangular hull standing out against the background noise.

This result demonstrates that the Poole Manifold can support the spontaneous formation and long-term stability of large-scale, coherent geometric structures that resemble engineered spacecraft.

4 Immortal Memory

The Poole Manifold supports memory that remains stable indefinitely without external refresh or correction. This is realised as the Immortal Latch (V207), a one-bit cell constructed from a pair of cross-coupled macro-orbits. Once set, kinetic mass circulates continuously around the closed loop. Under repeated high-amplitude noise bursts, the latch consistently restores its original state, demonstrating strong topological protection.

5 Self-Healing Logic

The model further exhibits structures that actively repair their own wiring by absorbing and redirecting incoming kinetic mass. In the Immortal Latch, a hostile strike that disrupts part of the circulating loop is automatically repaired. The prime-resonance mechanism concentrates repair material, while the gauge coupling steers the kinetic mass to the damaged region. Telemetry confirms that the structure not only survives but utilises the disruptive energy to restore and strengthen the waveguide.

6 Emergent Cosmology from the Same Rules

The identical prime-resonance sharpening and gauge coupling that produce computation also generate an expanding lattice with cosmological properties. When operated as an FLRW-style expanding lattice, macro-orbits act as discrete mass concentrations and the gauge term induces distance-dependent forces. A sustained succession flux $\Phi \approx 0.3095$ emerges naturally and drives accelerated expansion.

The effective equation of state is parametrised in CPL form:

$$w(z) = -1 + \alpha\beta \frac{z}{1+z}.$$

MCMC fitting against the latest DESI BAO data yields best-fit parameters $\alpha = 0.7944 \pm 0.0079$, $\beta = 1.9869 \pm 0.0191$, $\Omega_m = 0.3039 \pm 0.0035$, and r_d scale $= 1.0997 \pm 0.0004$. The OTG model improves the χ^2 fit over standard ΛCDM by $\Delta\chi^2 \approx 243.6$.

Thus the Poole Manifold constitutes a unified discrete substrate in which computation and gravity-like physics emerge from the same minimal local rules.

7 Discussion

The Poole Manifold demonstrates that a very simple three-dimensional totalistic cellular automaton can simultaneously support universal computation, immortal memory, self-healing logic, self-replication, evolutionary dynamics, emergent wormhole structures, stable particle-like excitations, macroscopic starship-like hulls, and emergent cosmological behaviour. All phenomena arise from the B5–7/S5–9 rule augmented only by prime-resonance sharpening. The constructions are minimal, reproducible on GPU hardware, and require no additional mechanisms or fine-tuning.

Limitations remain. Current simulations use grids up to 640^3; larger-scale runs will be required for quantitative scaling studies. The model is classical; extensions toward quantum-like behaviour are ongoing. The gauge coupling produces realistic forces but has not yet been calibrated to the weak-field limit of General Relativity.

Nevertheless, the Poole Manifold provides a compelling example of a minimal discrete system capable of rich computational and physical behaviour.

8 Conclusion

The Poole Manifold is a three-dimensional totalistic cellular automaton governed by the B5–7/S5–9 rule with prime-resonance sharpening. From this minimal rule set there emerge universal computation, immortal memory, self-healing logic, self-replication, evolutionary ecosystems, emergent wormhole structures, stable particle-like excitations, macroscopic starship-like hulls, and an expanding lattice whose cosmological properties fit DESI BAO data better than standard ΛCDM.

These results demonstrate that meaningful computation and physical-like behaviour can arise from simple discrete processes. They further illustrate that independent researchers with modest GPU resources can explore three-dimensional cellular automata in a substantive way.

9 Code Availability

The complete PyTorch implementation, including the V207 Immortal Latch, V140 Full Adder verification suite, V54 8-Bit Parallel Array, V67 OpCode Multiplexer, Trial 317 orthogonal blowout gates, Trial 318 logic expansion, Sci-Fi Test 1 self-replicating probe, Sci-Fi Test 2 self-evolving ecosystem, Trial 458 Einstein-Rosen Bridge, Particle Spectrum test, Starship hull simulation, ALU constructions, and all cosmological lattice scripts, is available at:

`https://github.com/rookepoole/SVP-OTG-Poole-Manifold-tests`